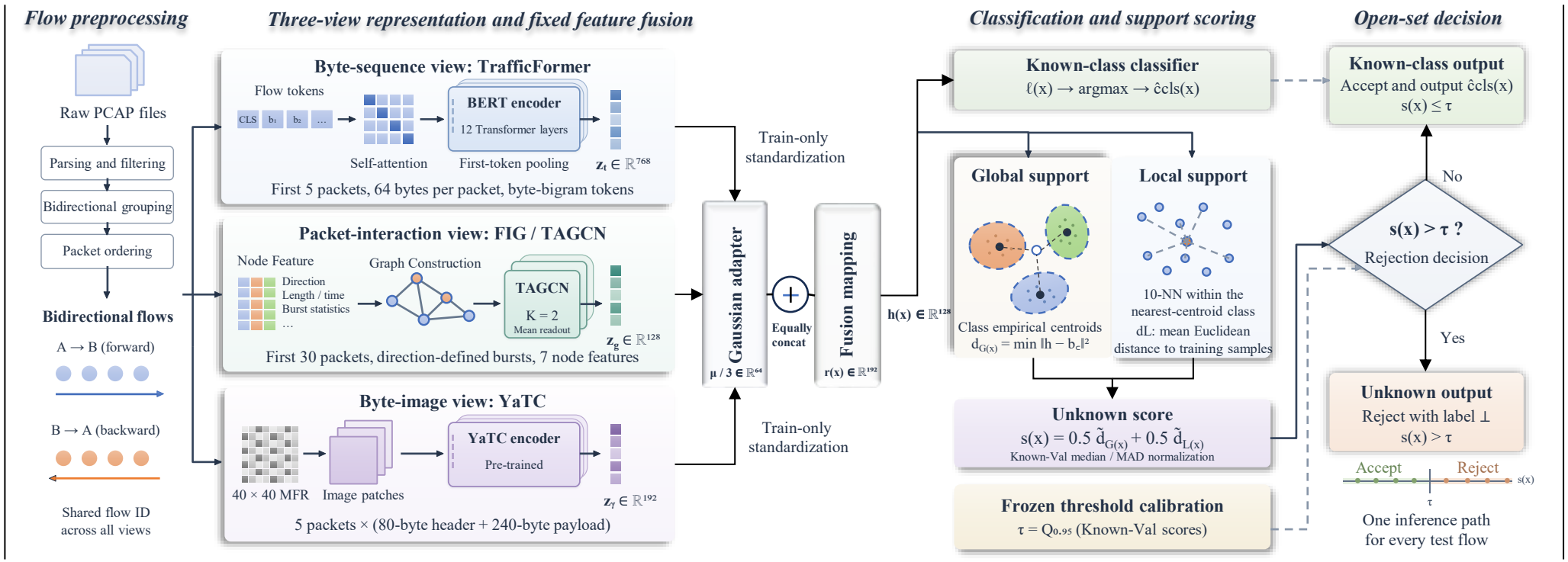

Flow preprocessing
Raw PCAP files
Parsing and filtering
Bidirectional grouping
Packet ordering
Bidirectional flows
A → B (forward)
B → A (backward)
Shared flow ID across all views
Three-view representation and fixed feature fusion
Byte-sequence view: TrafficFormer
Flow tokens
CLS
b₁
b₂
…
BERT encoder
12 Transformer layers
Self-attention
First-token pooling
$z_t \in \mathbb{R}^{768}$
First 5 packets, 64 bytes per packet, byte-bigram tokens
Packet-interaction view: FIG / TAGCN
Node Feature
Direction
Length / time
Burst statistics
…
Graph Construction
TAGCN
K = 2
Mean readout
$z_g \in \mathbb{R}^{128}$
First 30 packets, direction-defined bursts, 7 node features
Byte-image view: YaTC
40 × 40 MFR
Image patches
YaTC encoder
Pre-trained
$z_\gamma \in \mathbb{R}^{192}$
5 packets × (80-byte header + 240-byte payload)
Train-only standardization
Gaussian adapter
$\mu / 3 \in \mathbb{R}^{64}$
Equally concat
Fusion mapping
$r(x) \in \mathbb{R}^{192}$
$h(x) \in \mathbb{R}^{128}$
Train-only standardization
Classification and support scoring
Known-class classifier
$\ell(x) \rightarrow \text{argmax} \rightarrow \hat{c}\text{cls}(x)$
Global support
Class empirical centroids
$d_{G(x)} = \min \|h - b_c\|^2$
Local support
10-NN within the nearest-centroid class
dL: mean Euclidean distance to training samples
Unknown score
$s(x) = 0.5\, \tilde{d}_{G(x)} + 0.5\, \tilde{d}_{L(x)}$
Known-Val median / MAD normalization
Frozen threshold calibration
$\tau = Q_{0.95}$ (Known-Val scores)
Open-set decision
Known-class output
Accept and output $\hat{c}\text{cls}(x)$
$s(x) \le \tau$
No
$s(x) > \tau$?
Rejection decision
Yes
Unknown output
Reject with label ⊥
$s(x) > \tau$
Accept
Reject
s(x)
τ
One inference path for every test flow